

# NETWORK SECURITY

## Complete 6-Month Professional Training Program

*Theory + Hands-On Labs + Capstone Project*

Duration: 24 Weeks | 5 Days/Week | 6 Hours/Day | ~720 Training Hours

Designed for Corporate Training Programs & College Curricula

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## Course Overview

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This program takes a learner from networking basics to independently executing and reporting a full red-team-vs-blue-team engagement. Every week pairs theory with a hands-on lab so students build muscle memory, not just notes. The course is delivered in an isolated, instructor-controlled lab environment; all offensive techniques are practiced only against systems the student owns or is explicitly authorized to test (Metasploitable, DVWA, own lab AD/AP, etc.), and every offensive week is paired with the corresponding defensive/reporting skill.

### Who This Course Is For

- Fresh graduates (B.Tech/BCA/MCA/Diploma) targeting a career in cybersecurity or networking
- Corporate IT/network teams needing a structured security upskilling track
- System/network administrators moving into a security-focused role
- Individuals preparing for CompTIA Network+, Security+, CEH, or CCNA Security certifications

### Prerequisites

- Basic computer literacy and comfort using a command line
- A laptop with minimum 8GB RAM (16GB recommended) to run lab virtual machines
- No prior networking/security experience required - Month 1 starts from zero

### Daily Structure (6 Hours)

| Session              | Duration | Focus                                                           |
|----------------------|----------|-----------------------------------------------------------------|
| Theory Session       | 2.5 hrs  | Concepts, whiteboard walkthroughs, real-world case studies, Q&A |
| Break                | 0.5 hr   | Lunch/rest break                                                |
| Hands-on Lab Session | 3 hrs    | Guided lab + independent practice on the day's tool/technique   |

### Learning Outcomes - By End of Program, Students Will Be Able To

- Design, configure, and troubleshoot switched/routed networks with VLANs, NAT, and dynamic routing
- Administer Linux and Windows Server environments relevant to security operations
- Apply cryptography, PKI, and identity/access management correctly in real systems
- Deploy and tune firewalls, IDS/IPS, and VPNs in a live topology
- Conduct an authorized penetration test end-to-end - recon, scanning, exploitation, reporting
- Operate SIEM tooling (Splunk/ELK), triage alerts, and hunt for threats using MITRE ATT&CK
- Execute the full NIST incident response lifecycle including basic disk/memory forensics
- Secure cloud (AWS/Azure), wireless, and IoT environments
- Independently plan, execute, and professionally report a full security engagement

## Program Structure at a Glance

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| Month | Weeks | Focus Area                                              |
|-------|-------|---------------------------------------------------------|
| 1     | 1-4   | Networking Fundamentals & System Administration         |
| 2     | 5-8   | Core Security Concepts, Cryptography & Governance       |
| 3     | 9-12  | Network Security Devices & Architecture                 |
| 4     | 13-16 | Ethical Hacking & Penetration Testing                   |
| 5     | 17-20 | SOC Operations, Threat Intelligence & Red/Blue Team Ops |
| 6     | 21-24 | Cloud Security, Wireless/IoT, Capstone & Career Prep    |

## Month 1: Networking Fundamentals & System Administration

### Week 1: Networking Foundations

| Day        | Theory (approx. 2.5 hrs)                                                                                            | Hands-on Lab / Practical (approx. 3.5 hrs)                                 |
|------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Mon</b> | Introduction to Networks, History of Internet, Networking career paths, Course roadmap & lab setup                  | Install VMware/VirtualBox, set up lab VMs (Kali, Windows Server, Ubuntu)   |
| <b>Tue</b> | OSI Model - all 7 layers explained with real protocol mapping                                                       | Wireshark install; capture and identify OSI layers in live traffic         |
| <b>Wed</b> | TCP/IP Model, Encapsulation/De-encapsulation, comparison with OSI                                                   | Wireshark: trace TCP 3-way handshake, analyze packet headers               |
| <b>Thu</b> | Network devices - Hub, Switch, Router, Bridge, Access Point, transmission media, cabling standards (Cat5e/6, fiber) | Cable crimping (straight/cross), Packet Tracer: build a basic LAN topology |
| <b>Fri</b> | Ethernet fundamentals, MAC addressing, collision/broadcast domains, duplex modes                                    | Packet Tracer: switch configuration, MAC table inspection, Week 1 quiz     |

**Tools/Platforms:** Cisco Packet Tracer, Wireshark

**Week Outcome:** *Students can explain the OSI/TCP-IP model, identify network devices, and capture/analyze basic traffic in Wireshark.*

### Week 2: IP Addressing, Subnetting & Routing Basics

| Day        | Theory (approx. 2.5 hrs)                                           | Hands-on Lab / Practical (approx. 3.5 hrs)                             |
|------------|--------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Mon</b> | IPv4 addressing, classful vs classless, private/public IP ranges   | IP addressing exercises, subnet calculator practice                    |
| <b>Tue</b> | Subnetting deep-dive - CIDR, VLSM, subnet math tricks              | 20+ subnetting problems solved manually and verified                   |
| <b>Wed</b> | IPv6 addressing fundamentals & transition mechanisms               | Configure IPv6 addressing in Packet Tracer                             |
| <b>Thu</b> | Routing concepts - static routing, routing tables, default gateway | Packet Tracer: configure static routes across 3+ routers               |
| <b>Fri</b> | Intro to dynamic routing - RIP, OSPF, EIGRP overview               | Packet Tracer: configure basic OSPF between routers, Week 2 assessment |

**Tools/Platforms:** Cisco Packet Tracer, ipcalc

**Week Outcome:** *Students can subnet any given network (VLSM/CIDR) and configure static routing between multiple networks.*

**Week 3: Switching, VLANs, DHCP/DNS, NAT**

| Day        | Theory (approx. 2.5 hrs)                                   | Hands-on Lab / Practical (approx. 3.5 hrs)                            |
|------------|------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Mon</b> | Switching concepts, STP basics, port security fundamentals | Configure port security & STP in Packet Tracer                        |
| <b>Tue</b> | VLAN concepts, trunking (802.1Q), VTP                      | Configure VLANs + trunk ports across 2 switches                       |
| <b>Wed</b> | Inter-VLAN routing, Router-on-a-stick, Layer 3 switching   | Full inter-VLAN routing lab build                                     |
| <b>Thu</b> | DHCP working, DORA process, DNS hierarchy & record types   | Configure DHCP server & DNS resolution in lab topology                |
| <b>Fri</b> | NAT/PAT concepts, port forwarding                          | Configure NAT/PAT on router, verify with Wireshark; Week 3 assessment |

**Tools/Platforms:** Cisco Packet Tracer, Wireshark

**Week Outcome:** *Students can design a segmented switched network with VLANs, inter-VLAN routing, DHCP, DNS and NAT configured end-to-end.*

**Week 4: Linux & Windows Administration for Security Professionals**

| Day        | Theory (approx. 2.5 hrs)                                                    | Hands-on Lab / Practical (approx. 3.5 hrs)                 |
|------------|-----------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Mon</b> | Linux fundamentals - filesystem hierarchy, shell basics, package management | Kali/Ubuntu terminal bootcamp - navigation, file ops       |
| <b>Tue</b> | Linux users, groups, permissions (chmod/chown), sudoers, processes          | Create users/groups, permission hardening exercise         |
| <b>Wed</b> | Linux networking commands, services & systemd, log locations (/var/log)     | Configure a service, inspect and filter logs with grep/awk |
| <b>Thu</b> | Windows Server basics - Active Directory concepts, Group Policy overview    | Install AD DS, create OU/users/GPO on Windows Server       |
| <b>Fri</b> | PowerShell fundamentals for security automation                             | PowerShell scripting exercise; Month 1 practical exam      |

**Tools/Platforms:** Kali Linux, Ubuntu Server, Windows Server 2022, PowerShell

**Week Outcome:** *Students are comfortable administering Linux/Windows systems, users, permissions, and services from the command line - a prerequisite for every later security lab.*

## Month 2: Core Security Concepts, Cryptography & Governance

### Week 5: Information Security Fundamentals

| Day        | Theory (approx. 2.5 hrs)                                                               | Hands-on Lab / Practical (approx. 3.5 hrs)                                       |
|------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Mon</b> | CIA Triad, AAA framework, security terminology (threat, vulnerability, risk, exploit)  | Risk identification workshop on sample case studies                              |
| <b>Tue</b> | Types of attackers, motives, Cyber Kill Chain, MITRE ATT&CK framework                  | Map 3 historical breaches to Kill Chain/ATT&CK stages                            |
| <b>Wed</b> | Malware types - virus, worm, trojan, ransomware, rootkit, spyware                      | Safe malware behavior analysis in isolated sandbox (Any.Run/theZoo - controlled) |
| <b>Thu</b> | Social engineering - phishing, pretexting, baiting, vishing                            | Build a phishing-awareness simulation email (defensive training use)             |
| <b>Fri</b> | Security architecture principles - defense in depth, least privilege, zero trust intro | Design a defense-in-depth diagram for a sample org; Week 5 quiz                  |

**Tools/Platforms:** MITRE ATT&CK Navigator, Cyber Kill Chain resources

**Week Outcome:** *Students can classify threats/vulnerabilities/risks and map real attacks to the Cyber Kill Chain and MITRE ATT&CK framework.*

### Week 6: Cryptography I - Foundations

| Day        | Theory (approx. 2.5 hrs)                                                                | Hands-on Lab / Practical (approx. 3.5 hrs)                                                                  |
|------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Mon</b> | Cryptography history, goals (confidentiality, integrity, authenticity, non-repudiation) | CyberChef - encode/decode classical ciphers                                                                 |
| <b>Tue</b> | Symmetric encryption - DES, 3DES, AES, block/stream ciphers, modes (CBC, GCM)           | OpenSSL: AES file encryption/decryption exercises                                                           |
| <b>Wed</b> | Asymmetric encryption - RSA, Diffie-Hellman, ECC fundamentals                           | OpenSSL: generate RSA key pairs, encrypt/decrypt messages                                                   |
| <b>Thu</b> | Hashing algorithms - MD5, SHA family, HMAC, salting                                     | Hash generation & integrity verification lab; password hash cracking with Hashcat (educational, own hashes) |
| <b>Fri</b> | Digital signatures, hybrid encryption (how TLS actually works)                          | Sign & verify a file using GPG; Week 6 assessment                                                           |

**Tools/Platforms:** OpenSSL, CyberChef, GPG

**Week Outcome:** *Students can perform symmetric/asymmetric encryption and hashing operations using OpenSSL and explain the math intuition behind each.*

**Week 7: Cryptography II, PKI & Identity Management**

| Day        | Theory (approx. 2.5 hrs)                                                           | Hands-on Lab / Practical (approx. 3.5 hrs)                                           |
|------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Mon</b> | PKI architecture - CA, RA, certificate lifecycle, X.509 structure                  | Build a private CA with OpenSSL                                                      |
| <b>Tue</b> | SSL/TLS handshake deep-dive, certificate chains, common TLS misconfigurations      | Issue and install a self-signed cert on Apache/Nginx, inspect handshake in Wireshark |
| <b>Wed</b> | Authentication factors, MFA/2FA, biometrics, SSO, Kerberos & LDAP basics           | Configure MFA (TOTP) on a demo app with Keycloak                                     |
| <b>Thu</b> | IAM concepts - RBAC, ABAC, Privileged Access Management (PAM)                      | Design an RBAC matrix for a sample organization; configure AD security groups        |
| <b>Fri</b> | Password policies, credential attacks (brute force, credential stuffing, spraying) | Password policy hardening lab in Windows Server; Week 7 assessment                   |

**Tools/Platforms:** OpenSSL, Microsoft AD CS, Okta/Keycloak (community), Google Authenticator

**Week Outcome:** *Students can stand up a mini Certificate Authority, issue certificates, and configure MFA on a test application.*

**Week 8: Governance, Risk, Compliance & Malware Fundamentals**

| Day        | Theory (approx. 2.5 hrs)                                                            | Hands-on Lab / Practical (approx. 3.5 hrs)                          |
|------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Mon</b> | Security policies, standards, procedures, guidelines; BCP/DR concepts               | Draft a sample Acceptable Use Policy                                |
| <b>Tue</b> | ISO 27001, NIST CSF, PCI-DSS, GDPR/DPDP Act overview & applicability in India       | Compliance gap-analysis exercise on a sample company profile        |
| <b>Wed</b> | Risk management lifecycle - identification, assessment, mitigation, residual risk   | Build a risk register with likelihood/impact scoring                |
| <b>Thu</b> | Malware analysis fundamentals - static vs dynamic analysis, sandboxing safety rules | PeStudio + VirusTotal static analysis of a sample benign/known file |
| <b>Fri</b> | Security awareness training design, incident reporting culture                      | Month 2 practical + written exam                                    |

**Tools/Platforms:** NIST CSF, ISO 27001 checklist templates, PeStudio, VirusTotal

**Week Outcome:** *Students understand major compliance frameworks and can perform a basic static malware triage safely.*



## Month 3: Network Security Devices & Architecture

### Week 9: Firewalls

| Day        | Theory (approx. 2.5 hrs)                                            | Hands-on Lab / Practical (approx. 3.5 hrs)                               |
|------------|---------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Mon</b> | Firewall types - packet filtering, stateful, proxy, NGFW; ACL logic | iptables: build INPUT/OUTPUT/FORWARD rule sets on Ubuntu                 |
| <b>Tue</b> | pfSense architecture & deployment models                            | Deploy pfSense VM as network gateway                                     |
| <b>Wed</b> | Rule design best practices, default-deny, rule ordering             | pfSense: create allow/deny rules, NAT, test with nmap from client        |
| <b>Thu</b> | NGFW features - deep packet inspection, application-aware filtering | Cisco ASA basic config in Packet Tracer                                  |
| <b>Fri</b> | Firewall logging & alerting fundamentals                            | pfSense logging review + firewall rule audit exercise; Week 9 assessment |

**Tools/Platforms:** pfSense, iptables, Cisco ASA (Packet Tracer), Windows Firewall Advanced Security

**Week Outcome:** *Students can deploy and configure a firewall with custom rule sets, NAT, and logging in a live lab topology.*

### Week 10: IDS / IPS

| Day        | Theory (approx. 2.5 hrs)                               | Hands-on Lab / Practical (approx. 3.5 hrs)                           |
|------------|--------------------------------------------------------|----------------------------------------------------------------------|
| <b>Mon</b> | IDS vs IPS, signature-based vs anomaly-based detection | Install Snort on Ubuntu                                              |
| <b>Tue</b> | Snort rule syntax and structure                        | Write custom Snort rules to detect ICMP/port scans                   |
| <b>Wed</b> | Suricata architecture, comparison with Snort           | Deploy Suricata, test against Nmap scan from attacker VM             |
| <b>Thu</b> | Security Onion platform overview - NSM concepts        | Deploy Security Onion in lab, explore Kibana/Zeek data               |
| <b>Fri</b> | Tuning IDS/IPS to reduce false positives               | Simulated attack detection exercise, rule tuning; Week 10 assessment |

**Tools/Platforms:** Snort, Suricata, Security Onion

**Week Outcome:** *Students can write custom detection rules and detect a simulated attack (port scan, brute force) in real time.*

**Week 11: VPNs & Secure Remote Access**

| Day        | Theory (approx. 2.5 hrs)                                              | Hands-on Lab / Practical (approx. 3.5 hrs)                 |
|------------|-----------------------------------------------------------------------|------------------------------------------------------------|
| <b>Mon</b> | VPN concepts, tunneling protocols, IPSec (AH/ESP), IKE phases         | pfSense: configure site-to-site IPSec VPN between two labs |
| <b>Tue</b> | SSL VPN vs IPSec VPN, use cases                                       | Deploy OpenVPN server + client connectivity test           |
| <b>Wed</b> | Modern VPN protocols - WireGuard architecture                         | Deploy WireGuard VPN, benchmark vs OpenVPN                 |
| <b>Thu</b> | Proxy servers - forward/reverse proxy, load balancers & security role | Configure Squid forward proxy with content filtering       |
| <b>Fri</b> | Remote access security - Zero Trust Network Access (ZTNA) intro       | VPN troubleshooting scenarios; Week 11 assessment          |

**Tools/Platforms:** OpenVPN, WireGuard, pfSense IPSec, Squid Proxy

**Week Outcome:** *Students can configure and troubleshoot a site-to-site IPSec VPN and a client-based SSL VPN.*

**Week 12: Network Architecture, Segmentation & Zero Trust**

| Day        | Theory (approx. 2.5 hrs)                             | Hands-on Lab / Practical (approx. 3.5 hrs)                                                               |
|------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Mon</b> | Network segmentation strategies, DMZ design          | Build a 3-tier network (DMZ/internal/management) in Packet Tracer                                        |
| <b>Wed</b> | Zero Trust Architecture (NIST SP 800-207) principles | Map Zero Trust controls onto the built topology                                                          |
| <b>Tue</b> | SDN & network virtualization security implications   | Explore VXLAN/overlay concepts (conceptual + Packet Tracer demo)                                         |
| <b>Thu</b> | Network Access Control (NAC), 802.1X authentication  | Configure 802.1X port-based authentication demo                                                          |
| <b>Fri</b> | Enterprise architecture case study review            | Month 3 capstone mini-project: design & document a secure network for a fictional company; presentations |

**Tools/Platforms:** Cisco Packet Tracer, draw.io, pfSense

**Week Outcome:** *Students can design a segmented enterprise network (DMZ + internal zones) applying Zero Trust principles end-to-end.*

## Month 4: Ethical Hacking & Penetration Testing

### Week 13: Reconnaissance & Scanning

| Day        | Theory (approx. 2.5 hrs)                                                                                            | Hands-on Lab / Practical (approx. 3.5 hrs)                             |
|------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Mon</b> | Pentest methodology (PTES/OSSTMM), rules of engagement, scoping, legal/ethical boundaries & authorization documents | Draft a sample Rules-of-Engagement / authorization letter              |
| <b>Tue</b> | Passive OSINT - WHOIS, DNS recon, Google dorking, Shodan                                                            | theHarvester + Shodan recon exercise on lab-authorized target          |
| <b>Wed</b> | Active recon - Nmap fundamentals, host discovery, port states                                                       | Nmap scans (SYN, UDP, version, OS detection) against Metasploitable VM |
| <b>Thu</b> | Advanced Nmap - NSE scripting, evasion techniques                                                                   | Custom NSE script scans, firewall-evasion scan lab                     |
| <b>Fri</b> | Enumeration - SMB, SNMP, NetBIOS, banner grabbing                                                                   | Enumerate services on Metasploitable2/3; Week 13 assessment            |

**Tools/Platforms:** Kali Linux, Nmap, theHarvester, Maltego, Shodan

**Week Outcome:** *Students can perform full passive + active recon and produce a scoped scan report on a lab target.*

### Week 14: Vulnerability Assessment

| Day        | Theory (approx. 2.5 hrs)                                              | Hands-on Lab / Practical (approx. 3.5 hrs)                       |
|------------|-----------------------------------------------------------------------|------------------------------------------------------------------|
| <b>Mon</b> | Vulnerability management lifecycle, CVE/CVSS/CWE explained            | Install Nessus Essentials in lab                                 |
| <b>Tue</b> | Scan types - unauthenticated vs authenticated, credentialed scanning  | Run authenticated scan on Windows/Linux lab VMs                  |
| <b>Wed</b> | OpenVAS/Greenbone architecture                                        | Deploy OpenVAS, scan Metasploitable, compare results with Nessus |
| <b>Thu</b> | Web-focused scanning with Nikto, false-positive triage                | Nikto scan on DVWA/lab web server, validate findings manually    |
| <b>Fri</b> | Report writing - executive summary vs technical findings, risk rating | Write a full VA report on lab findings; Week 14 assessment       |

**Tools/Platforms:** Nessus, OpenVAS/Greenbone, Nikto

**Week Outcome:** *Students can run a full authenticated vulnerability scan, interpret CVSS scores, and write a prioritized findings report.*

**Week 15: Exploitation & Post-Exploitation**

| Day        | Theory (approx. 2.5 hrs)                                                               | Hands-on Lab / Practical (approx. 3.5 hrs)                                      |
|------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Mon</b> | Exploitation concepts, exploit-DB, payload types (staged vs non-staged)                | Metasploit Framework fundamentals - msfconsole walkthrough                      |
| <b>Tue</b> | Exploiting known CVEs on Metasploitable2/3 (isolated lab only)                         | Gain initial shell access via Metasploit on lab VM                              |
| <b>Wed</b> | Windows privilege escalation techniques (misconfigurations, token abuse - conceptual)  | PowerUp/WinPEAS enumeration on lab Windows VM, escalate in controlled lab       |
| <b>Thu</b> | Linux privilege escalation techniques (SUID, cron, kernel - conceptual)                | LinPEAS enumeration & escalation on lab Linux VM                                |
| <b>Fri</b> | Post-exploitation - persistence concepts, pivoting, evidence handling & cleanup ethics | Document full attack chain as a professional pentest report; Week 15 assessment |

**Tools/Platforms:** Metasploit Framework, Metasploitable2/3, Windows/Linux privilege escalation scripts

**Week Outcome:** *Students can exploit a known vulnerable service end-to-end, escalate privileges, and maintain documented evidence per engagement.*

**Week 16: Web Application Penetration Testing**

| Day        | Theory (approx. 2.5 hrs)                                                     | Hands-on Lab / Practical (approx. 3.5 hrs)                 |
|------------|------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Mon</b> | OWASP Top 10 overview, web app architecture, HTTP fundamentals recap         | Burp Suite setup, proxy interception basics on DVWA        |
| <b>Tue</b> | Injection flaws - SQLi types, blind/error-based, mitigation                  | Manual + SQLmap-assisted SQLi lab on DVWA/bWAPP            |
| <b>Wed</b> | XSS (reflected/stored/DOM), CSRF concepts                                    | XSS payload lab, CSRF PoC on DVWA                          |
| <b>Thu</b> | Broken authentication, session management flaws, IDOR, SSRF                  | Burp Suite: session token analysis, IDOR exploitation lab  |
| <b>Fri</b> | Security misconfiguration, file upload vulnerabilities, business-logic flaws | Full mini web-app pentest + report; Month 4 practical exam |

**Tools/Platforms:** Burp Suite, OWASP ZAP, DVWA, bWAPP, SQLmap

**Week Outcome:** *Students can identify and safely demonstrate each OWASP Top 10 category on a lab application and produce a web-app pentest report.*

## Month 5: SOC Operations, Threat Intelligence & Red/Blue Team Ops

### Week 17: SOC Fundamentals & SIEM

| Day        | Theory (approx. 2.5 hrs)                                                           | Hands-on Lab / Practical (approx. 3.5 hrs)                             |
|------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Mon</b> | SOC structure - Tier 1/2/3 analyst roles, SOC workflows, shift handover practices  | Explore a sample SOC ticketing/alert workflow                          |
| <b>Tue</b> | SIEM concepts - log sources, normalization, correlation rules                      | Install Splunk Free, ingest Windows/Linux logs via Universal Forwarder |
| <b>Wed</b> | ELK Stack architecture as an alternative SIEM                                      | Deploy ELK stack, ship logs with Filebeat, build a Kibana dashboard    |
| <b>Thu</b> | Writing correlation rules & use cases (failed logins, lateral movement indicators) | Build 3 SPL/Kibana detection use-cases and trigger them                |
| <b>Fri</b> | Alert triage methodology, false-positive handling, escalation criteria             | Triage simulated alerts, document findings; Week 17 assessment         |

**Tools/Platforms:** Splunk (Free), ELK Stack (Elasticsearch/Logstash/Kibana), Security Onion

**Week Outcome:** *Students can ingest logs into a SIEM, build a correlation search/dashboard, and triage a generated alert.*

### Week 18: Threat Intelligence & Threat Hunting

| Day        | Theory (approx. 2.5 hrs)                                                       | Hands-on Lab / Practical (approx. 3.5 hrs)                                 |
|------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Mon</b> | Threat Intelligence lifecycle - strategic/tactical/operational/technical intel | Explore OTX/MISP feeds, extract sample IOCs                                |
| <b>Tue</b> | IOCs vs IOAs, STIX/TAXII basics                                                | Build an IOC watchlist from a public threat report (summarized, own words) |
| <b>Wed</b> | Threat hunting methodology - hypothesis-driven hunting                         | Hunt for a simulated lateral-movement pattern in lab SIEM logs             |
| <b>Thu</b> | MITRE ATT&CK deep-dive - tactics/techniques/sub-techniques mapping             | Map a full simulated intrusion to the ATT&CK Navigator                     |
| <b>Fri</b> | Threat modeling - STRIDE, attack trees                                         | Threat-model a sample application; Week 18 assessment                      |

**Tools/Platforms:** MISP, MITRE ATT&CK Navigator, VirusTotal, AlienVault OTX

**Week Outcome:** *Students can consume threat intel feeds, map IOCs to ATT&CK techniques, and conduct a structured proactive hunt in lab logs.*

**Week 19: Red Team Operations**

| Day        | Theory (approx. 2.5 hrs)                                                            | Hands-on Lab / Practical (approx. 3.5 hrs)                           |
|------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Mon</b> | Red teaming vs pentesting, engagement types, C2 framework concepts                  | Deploy Sliver/open-source C2 in isolated lab network                 |
| <b>Tue</b> | Active Directory attack surface overview                                            | Build a small AD lab (1 DC + 2 workstations)                         |
| <b>Wed</b> | AD enumeration concepts - BloodHound/SharpHound data model                          | Run BloodHound against lab AD, analyze attack paths                  |
| <b>Thu</b> | Credential attacks in AD (conceptual: Kerberoasting, pass-the-hash) & why they work | Controlled demonstration in isolated lab with instructor supervision |
| <b>Fri</b> | Red team reporting & responsible disclosure of findings to Blue Team                | Document a full red-team engagement summary; Week 19 assessment      |

**Tools/Platforms:** Cobalt Strike (trial/edu) or Sliver C2, BloodHound, Mimikatz (isolated lab), Active Directory lab

**Week Outcome:** *Students understand and can demonstrate, in an isolated lab only, an end-to-end AD attack chain and how a C2 framework operates.*

**Week 20: Blue Team Operations - Incident Response & Forensics**

| Day        | Theory (approx. 2.5 hrs)                                                                                         | Hands-on Lab / Practical (approx. 3.5 hrs)                                                                     |
|------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Mon</b> | Incident Response lifecycle (NIST 800-61) - Prep, Detection, Containment, Eradication, Recovery, Lessons Learned | Build an IR runbook template                                                                                   |
| <b>Tue</b> | Digital forensics fundamentals - chain of custody, evidence integrity, imaging                                   | Create a forensic disk image with FTK Imager                                                                   |
| <b>Wed</b> | Memory forensics concepts, artifact analysis                                                                     | Analyze a memory dump with Volatility on a sample image                                                        |
| <b>Thu</b> | Endpoint forensics & triage at scale                                                                             | Use Velociraptor/KAPE for rapid triage collection on lab endpoint                                              |
| <b>Fri</b> | Tabletop exercise design & communication during incidents                                                        | Full simulated incident: red team's Week 19 attack is investigated & reported by blue team; Week 20 assessment |

**Tools/Platforms:** Velociraptor, Autopsy, FTK Imager, KAPE, Wireshark

**Week Outcome:** *Students can execute the NIST IR lifecycle end-to-end on a simulated incident and produce disk/memory forensic evidence.*

## Month 6: Cloud Security, Wireless/IoT, Capstone & Career Prep

### Week 21: Cloud Security

| Day        | Theory (approx. 2.5 hrs)                                                     | Hands-on Lab / Practical (approx. 3.5 hrs)                                                      |
|------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Mon</b> | Cloud service/deployment models, shared responsibility model                 | AWS/Azure free-tier account setup, console walkthrough                                          |
| <b>Tue</b> | IAM in the cloud - roles, policies, least privilege                          | Configure IAM users/roles/policies in AWS                                                       |
| <b>Wed</b> | Storage & network security in the cloud (S3/Blob, VPC/VNet, Security Groups) | Secure an S3 bucket, configure VPC security groups, fix a deliberately-misconfigured lab bucket |
| <b>Thu</b> | Cloud misconfigurations - common real-world breach patterns                  | Run ScoutSuite/Prowler against the lab AWS account, review findings                             |
| <b>Fri</b> | Container & Kubernetes security basics                                       | Docker security basics lab (image scanning with Trivy); Week 21 assessment                      |

**Tools/Platforms:** AWS Free Tier, Azure Free Tier, ScoutSuite, Prowler

**Week Outcome:** *Students can secure a basic cloud deployment (IAM, storage, network) and run an automated cloud security posture scan.*

### Week 22: Wireless & IoT Security

| Day        | Theory (approx. 2.5 hrs)                                                                         | Hands-on Lab / Practical (approx. 3.5 hrs)                                         |
|------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Mon</b> | Wireless fundamentals - 802.11 standards, WPA2/WPA3, EAP methods                                 | Wireless adapter/monitor mode setup in Kali                                        |
| <b>Tue</b> | WLAN attack concepts - deauth, evil twin, handshake capture (own lab AP only)                    | Aircrack-ng suite: capture handshake on own isolated lab AP, crack with dictionary |
| <b>Wed</b> | Enterprise wireless security - 802.1X/RADIUS, WIPS                                               | Configure WPA2-Enterprise auth in lab                                              |
| <b>Thu</b> | IoT security fundamentals - attack surface, common weaknesses (default creds, insecure firmware) | Assess a lab IoT device/emulated firmware for basic misconfigurations              |
| <b>Fri</b> | Wireless & IoT hardening checklist                                                               | Harden lab AP + IoT device configuration; Week 22 assessment                       |

**Tools/Platforms:** Aircrack-ng suite, Wireshark, WiFi Pineapple (demo), Kali Linux

**Week Outcome:** *Students understand WLAN attack/defense mechanics and can harden a wireless network and assess a basic IoT device.*

**Week 23: Capstone Project - Full-Scale Simulation**

| Day        | Theory (approx. 2.5 hrs)                                         | Hands-on Lab / Practical (approx. 3.5 hrs)                                              |
|------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Mon</b> | Capstone briefing - scenario assigned, teams split into Red/Blue | Environment setup & scoping document                                                    |
| <b>Tue</b> | —                                                                | Red Team: recon + exploitation phase / Blue Team: monitoring setup in SIEM              |
| <b>Wed</b> | —                                                                | Red Team: privilege escalation + lateral movement / Blue Team: detection & alert triage |
| <b>Thu</b> | —                                                                | Blue Team: incident response, containment & forensic collection                         |
| <b>Fri</b> | Cross-team debrief methodology                                   | Joint retrospective; both teams finalize reports                                        |

**Tools/Platforms:** Entire lab stack (Kali, pfSense, Snort/Suricata, Splunk/ELK, AD lab, Metasploitable)

**Week Outcome:** *Students complete a full end-to-end engagement covering the entire course: attack, detect, respond, and report.*

**Week 24: Capstone Presentations, Certification Prep & Career Readiness**

| Day        | Theory (approx. 2.5 hrs)                                                                      | Hands-on Lab / Practical (approx. 3.5 hrs)                       |
|------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>Mon</b> | Capstone report finalization workshop                                                         | Peer review of reports                                           |
| <b>Tue</b> | Final capstone presentations (Batch 1) to panel                                               | Q&A, scoring rubric feedback                                     |
| <b>Wed</b> | Final capstone presentations (Batch 2) to panel                                               | Q&A, scoring rubric feedback                                     |
| <b>Thu</b> | Certification roadmap - Security+/CEH/CCNA Security/OSCP path guidance, full-length mock exam | Timed mock exam + review of weak areas                           |
| <b>Fri</b> | Resume building, LinkedIn/GitHub portfolio, mock technical interviews                         | 1:1 mock interviews; course closing & certificates of completion |

**Tools/Platforms:** CompTIA Security+/Network+ mock exams, CEH practice sets, resume/LinkedIn tools

**Week Outcome:** *Every student presents a complete capstone report and leaves with a certification study plan, portfolio, and interview readiness.*



## Assessment & Evaluation Plan

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| Component                  | Weight | Details                                                                            |
|----------------------------|--------|------------------------------------------------------------------------------------|
| Weekly Quizzes/Labs        | 20%    | Short quiz + graded lab checkpoint every week                                      |
| Monthly Practical Exams    | 25%    | Hands-on practical exam at the end of each month (Weeks 4, 8, 12, 16, 20)          |
| Lab Reports                | 15%    | Professional-quality write-ups for VA scan, web-app pentest, and IR investigation  |
| Capstone Project           | 30%    | Full Red vs Blue simulation (Weeks 23-24) with final report and panel presentation |
| Attendance & Participation | 10%    | Lab attendance, in-class exercises, peer collaboration                             |

### Certification Alignment

- Months 1-2 map to CompTIA Network+ objectives
- Months 2-3 map to CompTIA Security+ objectives
- Month 4 maps to CEH (Certified Ethical Hacker) domains
- Months 3 & 5 map to CCNA Security concepts
- Month 4-5 build the foundation for OSCP-style practical assessments

## Lab Environment Requirements

### Per-Student Lab VM Set

- Kali Linux (attacker workstation)
- Windows Server 2022 (Domain Controller / target)
- Windows 10/11 client VM
- Ubuntu Server (services/target)
- Metasploitable2 / Metasploitable3 (deliberately vulnerable target)
- pfSense (firewall/gateway appliance)
- DVWA / bWAPP (deliberately vulnerable web apps)

### Complete Tool Stack Used Across the Program

| Category                 | Tools                                                      |
|--------------------------|------------------------------------------------------------|
| Networking               | Cisco Packet Tracer, Wireshark                             |
| OS Administration        | Kali Linux, Ubuntu Server, Windows Server, PowerShell      |
| Cryptography/PKI         | OpenSSL, GPG, CyberChef, Microsoft AD CS                   |
| Firewalls/VPN            | pfSense, iptables, OpenVPN, WireGuard, Squid               |
| IDS/IPS/NSM              | Snort, Suricata, Security Onion                            |
| Recon/Scanning           | Nmap, theHarvester, Shodan, Maltego                        |
| Vulnerability Assessment | Nessus, OpenVAS/Greenbone, Nikto                           |
| Exploitation             | Metasploit Framework, LinPEAS/WinPEAS                      |
| Web App Testing          | Burp Suite, OWASP ZAP, SQLmap                              |
| SIEM/SOC                 | Splunk Free, ELK Stack, Kibana                             |
| Threat Intel             | MISP, MITRE ATT&CK Navigator, VirusTotal, AlienVault OTX   |
| Red Team                 | Sliver C2 / Cobalt Strike (trial), BloodHound              |
| Forensics/IR             | Velociraptor, Autopsy, FTK Imager, KAPE, Volatility        |
| Cloud Security           | AWS Free Tier, Azure Free Tier, ScoutSuite, Prowler, Trivy |
| Wireless                 | Aircrack-ng suite                                          |

*Note: All offensive/attack techniques throughout this program are practiced exclusively within isolated, instructor-controlled lab environments against systems built specifically for training (own VMs, deliberately vulnerable applications, own lab wireless AP). Students sign a lab usage agreement in Week 1 committing to use these skills only with proper authorization.*